

Md Mahabubur Rahman

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PROFILE

AI and data science researcher completing a PhD in Computing at Glasgow Caledonian University, carried out in collaboration with Siemens Mobility Limited. My work develops AI-driven predictive maintenance for safety-critical engineering assets, combining time-series and deep learning forecasting, reliability and risk analysis, multi-criteria decision support and explainable AI, and extends it into a deployable prototype with SQL Server, FastAPI services, Power BI dashboards and a locally hosted agentic-AI layer with human-in-the-loop controls. I also bring computer vision experience from MSc research on automated defect detection in advanced manufacturing, five published peer-reviewed papers and one accepted paper as first author, and teaching experience as a Graduate Teaching Assistant and Associate Fellow of the Higher Education Academy.

RESEARCH INTERESTS

AI-driven predictive maintenance; time-series analysis and deep sequence models; reliability, risk and uncertainty quantification; explainable and trustworthy AI; local LLM and agentic-AI systems for engineering decision support; multi-criteria decision analysis; computer vision for inspection and defect detection; data engineering, model deployment and MLOps-style workflows; applied AI for transport, manufacturing and asset management.

EDUCATION

Doctor of Philosophy (PhD), Computing

Jan 2024 – Present

Glasgow Caledonian University, Glasgow, United Kingdom

- Thesis: Railway rolling stock fleet data analytics towards reliability and performance prediction.
- Collaborative doctoral research with Siemens Mobility Limited on an electric multiple unit (EMU) train fleet, covering five critical subsystems: CCTV, passenger doors, autocoupler, traction converter and pantograph.
- **Research focus:** data-driven modelling, reliability analysis, predictive maintenance and maintenance decision optimisation for safety-critical railway systems, using real multi-year operational and maintenance data.
- **Methods:** Bi-GRU and LSTM sequence models; SARIMAX, ETS and hybrid ARIMA-ANN forecasting; FMEA, SVM and Markov-chain risk modelling; fault-tree analysis; AHP/TOPSIS/PROMETHEE II/ELECTRE; Monte Carlo sensitivity testing; XAI and agentic-AI deployment prototyping.

MSc Advanced Data Science (Distinction)

Sep 2022 – Dec 2023

Bangor University, Bangor, United Kingdom

- Dissertation: Using machine learning and computer vision approaches in automatic data analysis for advanced manufacturing.
- Built an image-based inspection pipeline for SEM micrographs of additively manufactured parts, training and comparing detection and segmentation models for unmelted particles, porosity and microcracks.
- **Tools and methods:** Python, Google Colab, LabelMe, Mask R-CNN, Faster R-CNN, RetinaNet and YOLOv8.

BSc Computer Science and Engineering (GPA 3.50 / 4.00)

Feb 2017 – Apr 2021

Bangladesh University of Business & Technology (BUBT), Bangladesh

- Thesis: Breast cancer classification using deep learning technique with mammogram image.
- Developed a convolutional neural network (CNN) on the mini-MIAS mammography dataset to classify images as normal or abnormal (Python, Google Colab).

RESEARCH EXPERIENCE

PhD Researcher in Computing

Jan 2024 – Present

Glasgow Caledonian University, in collaboration with Siemens Mobility Limited

- **Time-series analysis and predictive models:** aggregated event-based fault records into weekly and monthly time series and derived model inputs; developed Bidirectional GRU models in TensorFlow/PyTorch and compared SARIMAX, LSTM, CNN and ETS for reliability forecasting.
- **Transferable modelling across asset types:** applied one modelling framework across five subsystems with different failure behaviour, data volumes and data quality, adapting methods and features to each asset rather than building isolated one-off models.
- **Maintenance decision support:** converted predictive outputs into ranked maintenance recommendations using AHP, TOPSIS, PROMETHEE II and ELECTRE with reliability, risk and service-impact criteria; tested ranking stability with 1,000-trial Monte Carlo analysis ($\pm 20\%$ input variation).

- **Deployable API and rolling-model prototype:** built a SQL Server-backed predictive-maintenance workflow with FastAPI/Uvicorn REST endpoints, pyodbc data access, dynamic rail-period control, rolling Bi-GRU retraining in WSL/TensorFlow GPU, model-version checks and persisted forecast outputs for downstream applications.
- **Explainable AI:** implemented post-hoc local temporal perturbation analysis for Bi-GRU forecasts, including observed historical contribution importance and +1 failure sensitivity; stored XAI outputs in SQL and exposed them in Power BI for engineer-facing interpretation.
- **Local LLM and agentic AI:** integrated Ollama with Qwen2.5:7b as a local planner and grounded explanation layer; implemented approved-tool routing, SQL-grounded question answering, question-aware context filtering, causal-language checks, deterministic fallback responses and human approval gates for protected workflow stages.
- **Data pipelines and operational dashboards:** built SQL pipelines to clean, validate and integrate multi-year fault, maintenance and service-impact records, feeding Power BI dashboards for fleet/unit forecasts, XAI and maintenance priorities; handled missing, duplicated and inconsistently coded records.
- **Probabilistic risk and reliability modelling:** combined FMEA, support vector machines and discrete-time Markov chains for component risk states; performed fault-tree, MTBF, mission-failure probability and trend analyses, with delay/cancellation impact assessment.
- **Research proposal development:** contributed to preparing a collaborative academic-industry research funding proposal for a predictive maintenance project, including the methodology, work plan and Gantt chart.
- **Knowledge transfer and project delivery:** work with Siemens engineering, maintenance and data teams to define requirements, validate outputs and communicate results; document methods and deliver teaching/training support, while contributing to collaborative academic-industry project planning.

TEACHING AND ACADEMIC SUPPORT

Graduate Teaching Assistant (Part-time)

Sep 2024 – Present

Glasgow Caledonian University, Glasgow, United Kingdom

- Support teaching across Engineering Mathematics, Mathematics 1, Mathematics 2, MATLAB practical sessions and the Predictive Maintenance Lab, including weekly tutorials as tutor on Mathematics 1 (Mechanical Engineering).
- Help students with quantitative methods, problem solving and software-based analytical tasks; mentor an MSc dissertation project with dataset, research-direction and academic-writing support.

Data Science Demonstrator (Part-time)

Nov 2023 – Dec 2023

Bangor University, Bangor, United Kingdom

- Supported practical teaching and student learning in data science sessions.

PUBLICATIONS

1. Rahman, M.M., Alkali, B., Jain, A.K., Parrilla-Gutierrez, J., McNeil, C. and Nelson, J. (2025) 'The application of time series predictive maintenance model on rolling stock critical systems', *Advances in Mechanical Engineering*, 17(10). [doi:10.1177/16878132251384345](https://doi.org/10.1177/16878132251384345)
2. Rahman, M.M., Alkali, B., Jain, A.K., Parrilla-Gutierrez, J., McNeil, C. and Nelson, J. (2026) 'Risk-based predictive maintenance of EMU autocouplers: a hybrid AI modelling approach', *Seventh International Conference on Railway Technology: Research, Development and Maintenance*. Civil-Comp Press. [doi:10.4203/ccc.15.8.3](https://doi.org/10.4203/ccc.15.8.3)
3. Rahman, M.M., Alkali, B., Jain, A.K., Parrilla-Gutierrez, J., McNeil, C. and Nelson, J. (2026) 'Pantograph failure analysis and short-term forecasting for EMU fleet using hybrid and machine learning models', *Seventh International Conference on Railway Technology: Research, Development and Maintenance*. Civil-Comp Press. [doi:10.4203/ccc.15.8.1](https://doi.org/10.4203/ccc.15.8.1)
4. Rahman, M.M., Alkali, B., Jain, A.K., Parrilla-Gutierrez, J., McNeil, C. and Nelson, J. (2025) 'A data-driven approach for optimising rolling stock fleet using time series analysis and SARIMAX model', *13th IMA International Conference on Modelling in Industrial Maintenance and Reliability*. [doi:10.19124/ima.2025.01.34](https://doi.org/10.19124/ima.2025.01.34)
5. Rahman, M.M., Alkali, B., Jain, A.K., Parrilla-Gutierrez, J., McNeil, C. and Nelson, J. (2024) 'Predictive maintenance optimisation for CCTV systems in electric multiple unit trains using machine learning techniques', *Sixth International Conference on Railway Technology: Research, Development and Maintenance*. Civil-Comp Press. [doi:10.4203/ccc.7.8.11](https://doi.org/10.4203/ccc.7.8.11)
6. Rahman, M.M., Alkali, B., Jain, A.K., Parrilla-Gutierrez, J., McNeil, C. and Nelson, J. (2025) 'Reliability analysis of electric train fleet traction converters using the fault tree analysis approach', *10th International Conference on Product Lifecycle Modeling, Simulation and Synthesis*, Pilani, India. (Accepted for publication).

AWARDS AND RECOGNITION

1st Prize (Scotland Centre) and National Finalist, Future of Rail Presentation Competition

2026

IMechE Railway Division

- Awarded 1st Prize at the Scotland Centre heat for a presentation on AI-enabled predictive maintenance for railway rolling stock critical systems, and selected as a national finalist at the UK final.

CERTIFICATIONS, MEMBERSHIPS AND PROFESSIONAL DEVELOPMENT

- Associate Fellow of the Higher Education Academy (AFHEA), Advance HE — July 2025.
- Institution of Mechanical Engineers (IMechE) — Applicant member; Academic Review under assessment.
- Graduate Teaching Assistant Certificate, Glasgow Caledonian University — June 2024.
- Cyber Security Compliance Training, Glasgow Caledonian University — September 2024.

TECHNICAL SKILLS

- **Programming and data:** Python, SQL, Pandas, NumPy, SciPy, scikit-learn, statsmodels, Matplotlib, pyodbc; Microsoft SQL Server/LocalDB; Git and GitHub.
- **API, deployment and AI systems:** FastAPI, Uvicorn, REST APIs, SQL-backed inference pipelines, rolling model versioning, WSL/Linux, TensorFlow GPU workflows; Ollama and Qwen2.5:7b for local LLM planning and grounded agentic-AI interaction.
- **Explainable AI:** post-hoc local temporal perturbation sensitivity, observed-contribution importance, model-behaviour interpretation, SQL persistence and Power BI visualisation.
- **Deep learning and computer vision:** TensorFlow/Keras, PyTorch, OpenCV, LabelMe; RNN/GRU/Bi-GRU/LSTM; Mask R-CNN, Faster R-CNN, RetinaNet, YOLOv8.
- **Time series and reliability:** SARIMAX, ETS, hybrid ARIMA-ANN, FMEA, SVM, Markov chains, fault-tree analysis, MTBF, mission-failure probability, Monte Carlo sensitivity analysis.
- **Decision support and visualisation:** AHP, TOPSIS, PROMETHEE II, ELECTRE; Power BI dashboards and database-driven analytical pipelines.
- **Other tools:** Google Colab for GPU-enabled model development; MATLAB through teaching support and practical sessions; Microsoft Office.
- **Applied domains:** predictive maintenance, transport and railway systems, asset management, industrial data quality, advanced manufacturing defect detection and medical image classification.

REFERENCES

Available on request.